analytical chemistry

Article

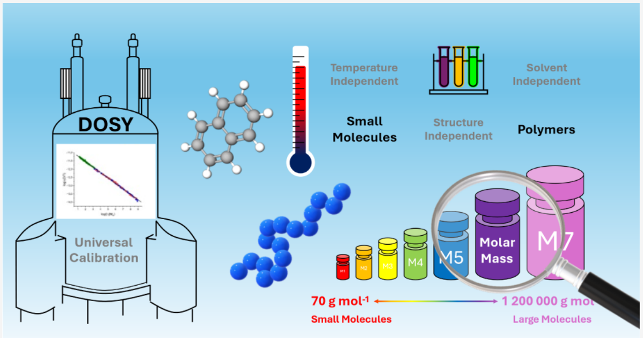
Molar Mass Determination for Small and Large Molecules Using Diffusion-Ordered Spectroscopy

Cite This:

ACCESS |

SI

ABSTRACT: The development of the most comprehensive universal calibration of molar mass dependences will be presented. For the first time, diffusion-ordered spectroscopy (DOSY) can now provide structure-, solvent-, and temperature-independent molar mass determinations for both small and large molecules. This fundamental theoretical approach provides only one single function which could perfectly describe all molar mass dependences. The new development using DOSY was tested on 477 diffusion coefficients representing altogether 56 molar mass dependences of 30 small molecules in 7 solvents, 5 different polymers in 10 solvents, and 11 temperature dependences of 2 polymers in 2 solvents, respectively. These samples cover a very large range of molar masses varying between 70 g/mol until 1 200 000 g/mol. The derived equation for the molar masses delivered a very good accuracy for all samples and might be one of the best tools for molar mass determinations.



■ INTRODUCTION

Diffusion-ordered spectroscopy was initially developed for the analysis of mixtures by separating the compounds according to their different diffusion coefficients. However, DOSY is also a very useful tool for the determination of sizes of molecules. By applying the Stokes−Einstein relation the hydrodynamic radius can be determined as follows:

$$r_h = \frac{k_B T}{6\pi\eta D} \quad (1)$$

where D is the diffusion coefficient, η the dynamic viscosity, k_B the Boltzmann constant and T the absolute temperature. was also used for describing shapes of molecule b including the shape correction factor f in the denominator. − This modification also allowed for the improved determination of diffusion coefficients of small molecules. Different expressions of the correction factors are published by Gierer and Wirtz and Chen and Chen. The Gierer and Wirtz setup was further adopted by Evans et al. by including an effective density and correlating t diffusion coefficients to the molar masses of the molecules. '

This approach was demonstrated for small molecules with molar masses up to 1500 g/mol. The group of Willard has shown the correlation of diffusion coefficients and formula weights by i an internal referencing and multinuclear DOSY NMR. −

One of the most common used approaches of DOSY is the power scaling concept. In this case, the diffusion coefficient can be directly correlat o the molar mass according to the Rouse-Zimm model '

$$D = cM^{-b} \quad (2)$$

where b is the Flory parameter and c is a constant. This scaling relation could be applied for smaller molecules and particularly for polymers. Whereas smaller molecules provide different Flory parameters, polymers of diluted solutions deliver Flory parameters mainly between 0.5 to 0.6. In particular, the works of Stalke et al. delivered very good results for the molar mass dependences of the diffusion coefficients of small molecules by assigning the molecules to three nt categories of shapes and using an external calibration. − In this case the Flory parameter depends on the shape. This method also works for complexes with heavy atoms by including a correction for heavy atoms.

In case of polymers, many different molar m s pendences of diffusion coefficients are published already. − Therefore, universal calibrations are more and more requested. The first publications are related to solvent-independent calibrations. Arrabal-Campos et al. used this method for polystyrene (PS),

Received: June 4, 2024
Revised: July 31, 2024
Accepted: August 26, 2024

Voorter et al. for PS ' and polyethylene oxide (PEO) and Ruzicka et al. for poly(methyl methacrylate). Tooley et al. applied this solvent-independent approach recently to a benchtop spectrometer of 80 MHz. Hiller and Grabe could already provide a structure- and solvent-independent calibration for three polymers and different solvents. This approach was covering only polymers within a range of 1 kDa to 100 kDa.

The idea of universal calibrations was originally developed for size exclusion chromatography (SEC). This is the most important method for determining molar masses of polymers by measuring polymer standards of a certain structure and different molar masses in dependence of the retention time. A so-called universal molar mass calibration can be obtained by combining the molar mass calibrations of different polymers in the same solvent to one uniform calibration curve. This principle is based on the Flory theory where the intrinsic viscosity $[\eta]$ is related to the hydrodynamic volume V_h and the molar mass M by

$$[\eta] = \frac{2.5 N_A V_h}{M} \tag{3}$$

where N_A is the Avogadro constant. The intrinsic viscosity can also be determined by the Mark–Houwink relation:

$$[\eta] = KM^a \tag{4}$$

where K and a are experimental parameters. The determination of the Mark–Houwink parameters K and a can be performed via SEC by using and . In this case, it can be proposed that two different polymers will have the same hydrodynamic volume at the s retention time. Consequently, can be written as: '

$$[\eta]_1 M_1 = [\eta]_2 M_2 \tag{5}$$

Including in , K_1 and a_1 are used as known parameters of a reference polymer in a certain solvent in order to directly calculate K_2 and a_2 of the other polymer.

However, it would be very useful to provide also a universal calibration for diffusion-ordered spectroscopy which is applicable to both small and large molecules in general.

EXPERIMENTAL SECTION

The polystyrene, poly(methyl methacrylate), polyisoprene, polyethylene oxide and poly(2-vinylpyridine) standards were produced by PSS GmbH (now Agilent Technologies). The molar masses of the polymers are shown in . The sample solutions were prepared at a concentration of 1 mg/mL with a total volume of 0.55 mL in 5 mm NMR tubes. The molar masses of the small molecules are summarized in .

The DOSY experiments were carried out on the 600 MHz Spectrometers AVANCE NEO and AVANCE III HD from Bruker BioSpin GmbH using helium cooled 5 mm cryoprobes. All DOSY experiments were performed with the Bruker dstebpgp3s pulse sequence consisting of the double stimulated echo for convection compensation and longitudinal eddy current delay (LED) with bipolar gradients for diffusion and three spoil gradient pulses. The gradient strengths were linearly changed between 3 and 98% of the maximum gradient strength. Altogether 32 gradient strengths were measured. The maximum gradient strength was calibrated with the doped water test sample providing a diffusion coefficient of $1.91 \cdot 10^{-9}$ m^2/s at 298.15 K. The adjusted parameters of the pulse sequence are the gradient pulse lengths and diffusion delays according to the molar mass of each polymer and the solvent viscosity. These data are summarized in . The acquisition time was 1.8–2.7 s (32 kb data), the spectrum window 10–15 ppm, relaxation delay 2.5 s, 16 scans per gradient strength and 16 dummy scans. The DOSY measurements were carried out minimum twice or three times at the calibrated temperatures in order to minimize experimental errors. Therefore, the integration of the best separated NMR signals allowed the determination of two to six diffusion coefficients per sample where the average value was used for the analysis. The temperatures of the NMR spectrometers were calibrated with the ethylene glycol test sample. An exponential window function with a line broadening of 1 Hz and zero filling of 64 kb data points in F2 and 1 kB in F1 was applied for the processing. The diffusion coefficients were calculated by using the Stejskal-Tanner equation with the corrected little and big delta delays (see).

RESULTS AND DISCUSSION

This work will show a more general universal calibration for diffusion coefficients that can be used for different structures,

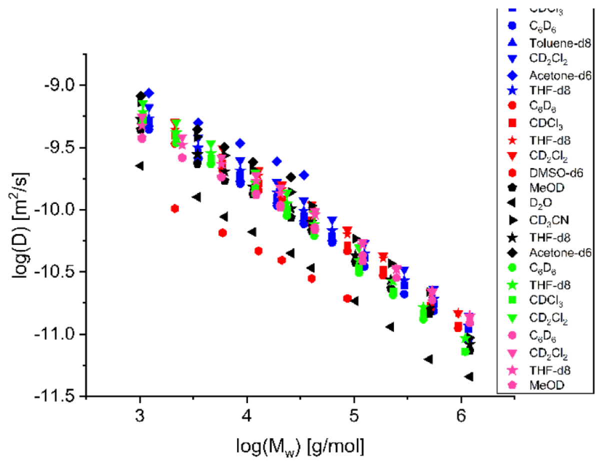


Figure 1. Diffusion coefficients vs molar masses of PS (blue), PMMA (red), PEO (black), PI (green) and P2VP (magenta) in different solvents indicated by the symbols.

different solvents as well as different temperatures and can be applied to small and large molecules, respectively. One uniform function will be derived for the molar mass determination with DOSY representing the structure-, solvent- and temperature-independent universal calibration.

The new method is based on the Flory theory by including the Mark–Houwink relation. This was first proposed by us and applied for three medium sized polymers in different solvents. Now this approach will be significantly extended by including temperature dependences, further polymers with much higher molar masses and in particular many small molecules in different solvents as well. Therefore, several orders of magnitudes of molar masses will be covered with this model.

The Mark–Houwink setup for DOSY will be achieved by including into for a spherical hydrodynamic volume and the combination with . This will result in the following expression:

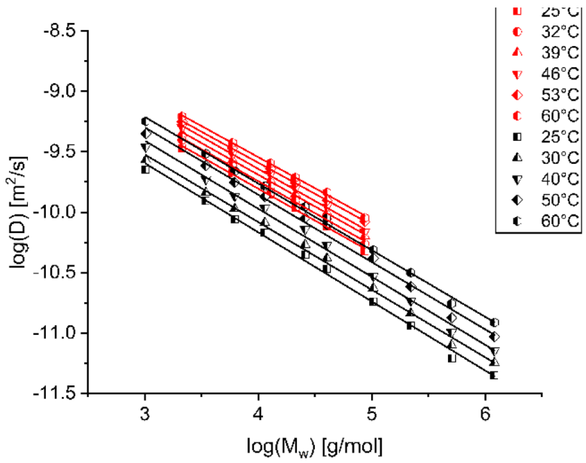


Figure 2. Temperature dependences of diffusion coefficients vs molar masses of PMMA in C_6D_6 (red) and PEO in D_2O (black). The different temperatures are given by the symbols.

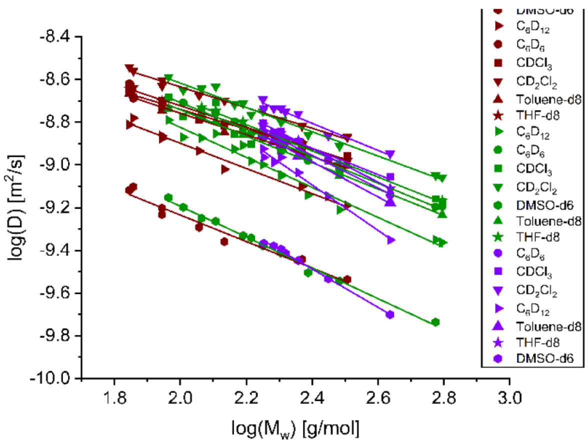


Figure 3. Diffusion coefficients vs molar masses of small molecules of different shapes: Compact spheres (brown), dissipated spheres and ellipsoids (green) and expanded discs (violet). The different solvents are indicated by the symbols. The data are reproduced from ref with permission from the Royal Society of Chemistry and from ref with permission from John Wiley and Sons.

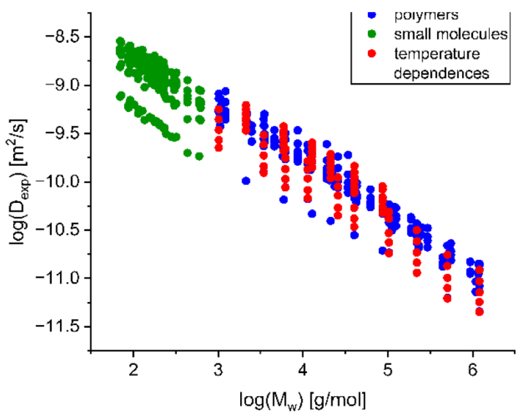


Figure 4. Diffusion coefficients vs molar masses: Altogether 477 diffusion coefficients are representing 56 molar mass dependences of 30 small molecules in 7 solvents (green circles), 5 polymers in 10 solvents (blue circles) and 11 temperatures dependences of two polymers in two solvents (red circles), respectively.

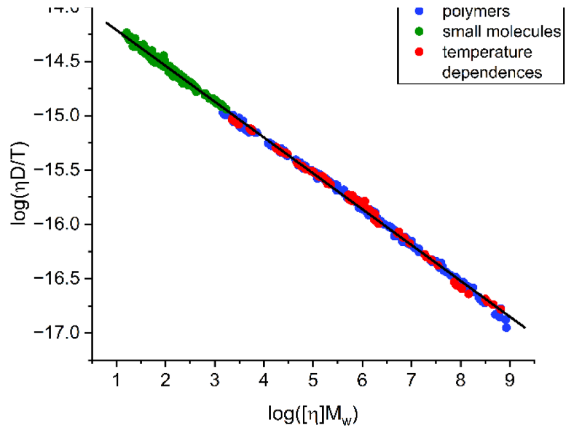


Figure 5. Diffusion coefficients corrected by the dynamic viscosity (given in Pa·s) and temperature vs molar masses corrected by the intrinsic viscosity with Mark−Houwink parameters for all small molecules (green circles), polymers (blue circles) and temperature dependences (red circles).

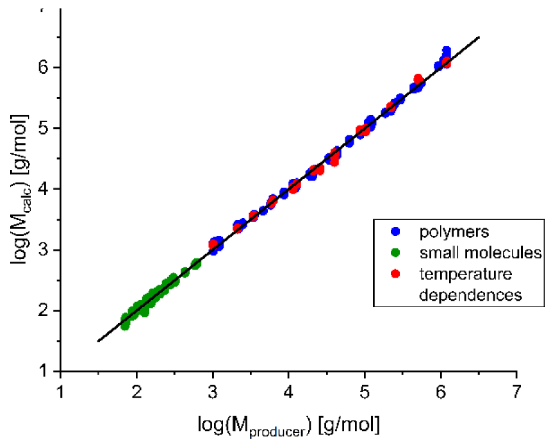


Figure 6. Calculated molar masses via with the parameters of Table 1 vs molar masses of the producer for all small molecules (green circles), polymers (blue circles) and temperature dependences (red circles). The black line represents the molar masses of the producer $M_{producer}$.

$$KM^{a+1} = \frac{10}{3}\pi N_A\left(\frac{k_B T}{6\pi\eta D}\right)^3 \tag{6}$$

correlates the diffusion coefficient to the intrinsic viscosity via the Mark−Houwink parameters a and K as well as to the dynamic viscosity and the variable temperature. This expression will now be used for the DOSY determination of Mark−Houwink parameters in different solvents and temperatures. The calculation of a and K can be carried out by correlating the data for two compounds where one compound presents the reference with known Mark−Houwink parameters K_1 and a_1. Consequently, the following ratio for two compounds can be derived from for the variable temperatures T_1 and T_2:

$$\frac{K_1M_1^{a_1+1}}{K_2M_2^{a_2+1}} = \left(\frac{T_1\eta_2D_2}{T_2\eta_1D_1}\right)^3 \tag{7}$$

The indices 1 and 2 refer to the reference compound with known K_1 and a_1 and index 2 to the compound with the

Table 1a. Mark-Houwink Parameters K and a Determined via DOSY for Small Molecules: Compact Spheres (CS), Dissipated Spheres and Ellipsoids (DSE) and Expanded Discs (ED)

Sample	THF-d8	C_6D_6	$CDCl_3$	CD_2Cl_2	Toluene-d8	DMSO-d6	C_6D_{12}
CS/K	0.066084	0.057471	0.098118	0.080843	0.041199	0.0052678	0.0093535
CS/a	0.52138	0.41888	0.4093	0.46991	0.53831	0.87833	0.78574
DSE/K	0.011091	0.0048527	0.018219	0.021919	0.0077827	0.00098646	0.0011657
DSE/a	0.85752	0.90806	0.75695	0.7266	0.87588	1.1805	1.1263
ED/K	0.00064684	0.00049889	0.0076064	0.0057885	0.00025459	0.00005603	0.00000268
ED/a	1.3592	1.3137	0.86571	0.91822	1.4893	1.7066	2.2114

Table 1b. Mark-Houwink Parameters K and a Determined via DOSY for Polystyrene (PS), Poly(Methyl Methacrylate) (PMMA), Polyisoprene (PI), Poly(2-Vinylpyridine) (P2VP), and Polyethylene Oxide (PEO)

Sample	THF-d8	C_6D_6	$CDCl_3$	CD_2Cl_2	Toluene-d8	Acetone-d6	DMSO-d6	CD_3CN	CD_3OD	D_2O
PS/K	0.0141	0.013427	0.013156	0.013431	0.024735	0.17932	-	-	-	-
PS/a	0.7	0.70012	0.70599	0.70367	0.61201	0.37508	-	-	-	-
PMMA/K	0.0090994	0.0085214	0.0062933	0.0058216	-	-	0.18104	-	-	-
PMMA/a	0.706	0.72619	0.76996	0.78089	-	-	0.35152	-	-	-
PI/K	0.0044444	0.0032451	0.0027791	0.011177	-	-	-	-	-	-
PI/a	0.84314	0.86722	0.90044	0.73833	-	-	-	-	-	-
P2VP/K	0.075189	0.11387	-	0.043726	-	-	-	-	0.1813	-
P2VP/a	0.53351	0.48529	-	0.62069	-	-	-	-	0.4555	-
PEO/K	0.01239	-	-	-	-	0.17246	-	0.0087817	0.015555	0.018426
PEO/a	0.76068	-	-	-	-	0.44224	-	0.8072	0.72943	0.73577

[a]K and a for PS in THF are the reference parameters and are related to literature.

Table 1c. Mark-Houwink Parameters K and a Determined via DOSY for PMMA in C_6D_6 and PEO in D_2O in Dependence of the Temperature in Kelvin

Sample	298.15 K	305.15 K	312.15 K	319.15 K	326.15 K	333.15 K
PMMA/K	0.027883	0.023566	0.026273	0.021795	0.034996	0.03421
PMMA/a	0.59799	0.6087	0.59536	0.61545	0.56125	0.56265
Sample	**298.15 K**	**303.15 K**	**313.15 K**	**323.15 K**	**333.15 K**	
PEO/K	0.018434	0.021827	0.018352	0.018483	0.021185	
PEO/a	0.73571	0.69844	0.70899	0.69122	0.67387	

unknown parameters K_2 and a_2, respectively. In addition, the calculation of the Mark–Houwink parameters requires also the replacement of molar masses M_1 and M_2. This is achieved by including into . As the result the following relation of the diffusion coefficients of the two compounds at variable temperatures will be obtained:

$$\log(D_2) = A + B\log(D_1) \tag{8}$$

with

$$A = \frac{b_2}{3b_2 - a_2 - 1}\left\{\log\left(\frac{K_1\eta_1^3T_2^3}{K_2\eta_2^3T_1^3}\right) + \frac{a_1+1}{b_1}\log(c_1) - \frac{a_2+1}{b_2}\log(c_2)\right\}$$

$$B = \frac{b_2(3b_1 - a_1 - 1)}{b_1(3b_2 - a_2 - 1)}$$

now allows the determination of the Mark–Houwink parameters K_2 and a_2 of any structure, in any solvent and at any temperature on the basis of a reference compound and the molar mass dependences of the diffusion coefficients of both compounds. The parameters b_1, b_2, c_1 and c_2 will be obtained from the molar mass dependences of the diffusion coefficients with . The parameters K_1 and a_1 are used as the references related to PS in THF from ref (see). K_2 and a_2 are determined by fitting to the diffusion coefficients D_2 and D_1 of the two molar mass dependences. D_2 and D_1 have to refer always to the same M_w values. This fitting procedure finally provides K_2 and a_2.

It should be noted that the dynamic viscosity is also a temperature-dependent parameter and can be described by the following equation:

$$\log(\eta) = \frac{d}{T} + e \tag{9}$$

with d and e as constants. Viscosity data are given in .

Furthermore, the parameters K_2 and a_2 obtained with are only relevant for DOSY. These parameters were only determined for the universal calibration of DOSY and can differ from Mark–Houwink parameters measured with SEC or viscometry.

In any case, allows the determination of Mark–Houwink parameters of different structures, in different solvents and at different temperatures for DOSY. Furthermore, it will be the basis for producing the universal molar mass calibration which is structure-, solvent- and temperature-independent.

In order to create a universal calibration for DOSY, the molar mass dependences of five different polymers were measured in several solvents, the temperature dependences of two polymers were measured as well as DOSY data of small molecules in several solvents were taken from literature.

shows the molar mass dependences of the diffusion coefficients for polystyrene (PS), poly (methyl methacrylate) (PMMA), polyisoprene (PI), poly(2-vinylpyridine) (P2VP) and polyethylene oxide (PEO) in several solvents. Altogether 10 solvents were used. shows the temperature dependences of PMMA in C_6D_6 as well as PEO in D_2O.

shows the molar mass depen es of 30 small molecules of different shapes in 7 solvents. ' The experimental DOSY conditions as well as the details to the molar masses of the polymers and small molecules are given in the experimental section and . The individual molar mass dependences of the polymers and small molecules are shown in . In order to present all molar mass dependences together, shows the summery of – with all diffusion coefficients in dependence on the molar masses by indicating the three categories with different colors. In this case, the large range of molar masses between 70 g/mol to 1 200 000 g/mol could be covered and the broad distribution of the molar mass dependences of the diffusion coefficients is visible.

Due to the extended molar mass ranges of the polymers, these parameters have changed also for PS, PEO and PMMA in comparison to ref .

is demonstrating the high accuracy of the calculated diffusion data with in comparison to the experimental data. The average deviation of the experimental diffusion coefficients from the average calculated line is just 4.7%.

In order to generate the universal calibration curve for the structure-, solvent- and temperature-independent molar mass determinations, the relevant parameters of Table 1 and have to be used and the appropriate proportionality derived from has to be applied. In this case, the diffusion coefficients corrected by the dynamic viscosities and temperatures will be plotted as the function of the molar masses corrected by the intrinsic viscosities. The intrinsic viscosities are calculated with the Mark–Houwink parameters of Table 1. According to this proportionality is given by the following expression:

$$\log\left(\frac{\eta D}{T}\right) \sim \log([\eta]M_w) \tag{10}$$

is showing now this universal calibration. It perfectly demonstrates that all diffusion data can be described by one uniform function:

$$\log\left(\frac{\eta D}{T}\right) = -0.3297\log([\eta]M_w) - 13.8825 \tag{11}$$

where η is given in Pa·s. If η is given in mPa·s then the intercept has to be replaced by 10.8825.

The accuracy of this calibration method is excellent. provides a simple linear correlation of the corrected diffusion coefficients to the corrected molar masses with an average deviation of only 4.5%. This result indicates its general usage. It can unify all different applications of molar mass determinations of diffusion coefficients. In some cases, it even improves the accuracy as shown for dynamic viscosity corrected data of polymers. The main reason for the general usage and the high accuracy of in is the introduction of the intrinsic viscosity via the Mark–Houwink parameters. Due to the molar mass dependence of the intrinsic viscosity, the molar mass itself can be corrected. As the consequence one general calibration function could be obtained from describing all diffusion coefficients of very different structures and shapes of small molecules in different solvents, many different polymers covering several magnitudes of molar masses in different solvents as well as temperature dependences of different polymers.

Finally, this new universal calibration allows for the determination of the molar mass of any of the presented compounds from a single diffusion coefficient by including the Mark–Houwink parameters, the dynamic viscosity and the temperature in Kelvin. In this case, the calibration function of can be resolved to the molar mass M_w and is given by:

$$\log(M_w) = -\frac{1}{a+1}\{3.0333\log(\eta D/T) + \log(K) + 42.1101\} \tag{12}$$

where η is given in Pa·s. If η is given in mPa·s then the last constant has to be replaced by 33.0101.

can be considered as a universal molar mass determination for DOSY due to the fact that it can be applied to small molecules and large polymers and it is particularly structure-, solvent- and temperature-independent. It can be even expected that this equation can be applied to much more compounds as long as Mark–Houwink parameters and dynamic viscosities are determined at the used temperatures. The Mark–Houwink parameters can be easily determined with .

Nevertheless, it should be noted that and should only be used at NMR systems with calibrated pulsed field gradients and calibrated temperatures, convection compensated pulse sequences, and the same conditions reported in the experimental section. Following these conditions, the diffusion coefficients measured at different NMR spectrometers are expected to provide the same results.

The final proof of is shown in . This figure presents the molar masses M_{calc} calculated with for all 477 diffusion coefficients in comparison to the molar masses given by the manufacturer. The average deviation is only 8%. In case of the small molecules with 5.3% even less. These are excellent results and it seems that DOSY might be a powerful tool for the molar mass determination in general. The , provided these accuracies by using up to four digits for both the derived numbers in these equations and the calculated parameters shown in – and . Five digits were taken for the temperature dependences of the dynamic viscosities ( ). In case of the calculated parameters the evaluation of the number of digits was using scientific notation.

■ CONCLUSION

It could be shown that DOSY molar mass calibrations for small and large molecules can be unified by one universal calibration function which is structure-, solvent- and temperature-independent. The implementation of the intrinsic viscosity together with the hydrodynamic volume and the diffusion coefficient allows for the derivation of the universal calibration function. The corresponding Mark–Houwink parameters representing the intrinsic viscosity can be calculated via DOSY measurements only. This approach finally provides a linear correlation of the diffusion coefficients corrected by the dynamic viscosity and temperature versus the molar masses corrected by the intrinsic viscosity. By implementing the temperature into the calibration function, the new equation could be applied to DOSY measurements of very different structures, any solvents and any temperatures. The function was successfully tested on small molecules of different sizes and shapes in seven solvents, five different polymers with a large range of molar masses in ten solvents and on two polymers at several temperatures in two solvents. Altogether 477 diffusion coefficients could be perfectly described with the derived eq. In particular, this equation can be used to determine the molar mass of any of these samples by simply measuring the diffusion coefficient in the certain solvent and temperature. The average deviation of the molar mass was only 8%. If molar mass dependences of other compounds are of interest, can be used to determine the Mark–Houwink parameters with DOSY. However, it cannot be assumed that these Mark–Houwink parameters are the same as determined by SEC and viscometry. Only parameters determined by DOSY should be used with the derived equations. It is also important to mention that the derived equations require the described experimental DOSY conditions and sample preparations.

Anyway, it would be very useful to further extend the database of Mark–Houwink parameters by DOSY to other compounds, solvents and temperatures.

ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at .

Additional experimental details, materials, figures, and methods ( )

AUTHOR INFORMATION

Corresponding Author

Wolf Hiller – *Faculty of Chemistry and Chemical Biology, TU Dortmund University, Dortmund 44227, Germany;* ; Email:

Authors

Bastian Grabe – *Faculty of Chemistry and Chemical Biology, TU Dortmund University, Dortmund 44227, Germany;*

Jan Schonert – *Faculty of Chemistry and Chemical Biology, TU Dortmund University, Dortmund 44227, Germany*

Complete contact information is available at:

Author Contributions

The manuscript was written through contributions of all authors. All authors have given approval to the final version of the manuscript.

Notes

The authors declare no competing financial interest.

ACKNOWLEDGMENTS

We are grateful for funding the 600 MHz NMR spectrometer AVANCE NEO by the DFG (German Research Foundation: project number - 452669688) and the Ministry of Culture and Science of North Rhine-Westphalia. We thank Prof. Dietmar Stalke from the University of Göttingen for the permission to use the diffusion data of the small molecules.

REFERENCES

(1) Einstein, A.
(2) Perrin, F.
(3) Aguerre, R. J.; Gabitto, J. F.; Chirife, J.
(4) Macchioni, A.; Ciancaleoni, G.; Zuccaccia, C.; Zuccaccia, D.
(5) Gierer, A.; Wirtz, K.
(6) Chen, H. C.; Chen, S. H.
(7) Evans, R.; Deng, Z.; Rogerson, A. K.; McLachlan, A. S.; Richards, J. J.; Nilsson, M.; Morris, G. A.
(8) Evans, R.; Dal Poggetto, G.; Nilsson, M.; Morris, G. A.
(9) Li, D.; Kagan, G.; Hopson, R.; Williard, P. G.
(10) Kagan, G.; Li, W.; Hopson, R.; Williard, P. G.
(11) Kagan, G.; Li, W.; Hopson, R.; Williard, P. G.
(12) Guang, J.; Hopson, R.; Williard, P. G.
(13) Zimm, B. H.
(14) Rouse, P. E.
(15) Bachmann, S.; Neufeld, R.; Dzemski, M.; Stalke, D.
(16) Neufeld, R.; Stalke, D.
(17) Bachmann, S.; Gernert, B.; Stalke, D.
(18) Kreyenschmidt, A.-K.; John, M.; Stalke, D.
(19) Chen, A.; Wu, D.; Johnson, C. S.
(20) Håkansson, B.; Nydén, M.; Söderman, O.
(21) Viel, S.; Capitani, D.; Mannina, L.; Segre, A.
(22) Augé, S.; Schmit, P.-O.; Crutchfield, C. A.; Islam, M. T.; Harris, D. J.; Durand, E.; Clemancey, M.; Quoineaud, A.-A.; Lancelin, J.-M.; Prigent, Y.; et al.
(23) Kuz'mina, N. E.; Moiseev, S. V.; Krylov, V. I.; Yashkir, V. A.; Merkulov, V. A.
(24) Kuz'mina, N. E.; Moiseev, S. V.; Krylov, V. I.; Yashkir, V. A.; Merkulov, V. A.
(25) Lewinski, P.; Sosnowski, S.; Kazmierski, S.; Penczek, S.
(26) Li, W.; Chung, H.; Daeffler, C.; Johnson, J. A.; Grubbs, R. H.
(27) Chamignon, C.; Duret, D.; Charreyre, M.-T.; Favier, A.
(28) Williamson, N. H.; Nydén, M.; Röding, M.
(29) Cherifi, N.; Khoukh, A.; Benaboura, A.; Billon, L.
(30) Groves, P.
(31) Vrijsen, J. H.; Thomlinson, I. A.; Levere, M. E.; Lyall, C. L.; Davidson, M. G.; Hintermair, U.; Junkers, T.
(32) Hiller, W.
(33) Grabe, B.; Hiller, W.
(34) Grabe, B.; Hiller, W.
(35) Arrabal-Campos, F. M.; Oña-Burgos, P.; Fernández, I.

(36) Voorter, P.-J.; McKay, A.; Dai, J.; Paravagna, O.; Cameron, N. R.; Junkers, T.
(37) Silva, I. W. F.; McKay, A.; Sokolova, A.; Junkers, T.

(38) Ruzicka, E.; Pellechia, P.; Benicewicz, B. C.

(39) Tooley, O.; Pointer, W.; Radmall, R.; Hall, M.; Beyer, V.; Stakem, K.; Swift, T.; Town, J.; Junkers, T.; Wilson, P.; Lester, D.; Haddleton, D.
(40) Hiller, W.; Grabe, B.
(41) Flory, P. J. *Principles of Polymer Chemistry*; Cornell University Press: Ithaka, N. Y, 1953.
(42) Rudin, A.; Johnston, H. K.
(43) Grubisic, Z.; Rempp, P.; Benoit, H.

(44) Kraus, G.; Stacy, C. J.

(45) Jerschow, A.; Müller, N.

(46) Stejskal, E. O.; Tanner, J. E.

(47) Wagner, H. L.